

Data Compression and Source Coding

Scalar Quantization

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Scalar Quantization

Original purpose: to discretize “analog” values. For example

$$x \rightarrow \left\lfloor x + \frac{1}{2} \right\rfloor \quad (\text{round to the nearest integer})$$

Definition (Scalar quantizer)

An N -point *scalar quantizer* is a mapping

$$Q : \mathbb{R} \rightarrow \mathcal{C}$$

where $\mathcal{C} = \{y_1, y_2, \dots, y_N\} \subset \mathbb{R}$.

- $\mathcal{C}$ is called the *codebook* of Q .
- $y_1, \dots, y_N$ are the *output levels* (or output points, reproduction points, quantization points, etc).
- We always assume indexing such that $y_1 < y_2 < \dots < y_N$.

- The sets $R_i = \{x : Q(x) = y_i\}$, $i = 1, \dots, N$ are called the *quantizer cells*.
- Clearly

$$R_i \cap R_j = \emptyset \quad \text{if} \quad i \neq j$$

and

$$\bigcup_{i=1}^N R_i = \mathbb{R}$$

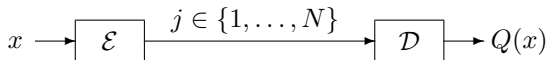
so that $R_1, \dots, R_N$ form a *partition* of $\mathbb{R}$.

- **Note:** Q is completely described by $\mathcal{C}$ and $\{R_1, \dots, R_N\}$ since

$$Q(x) = y_i \quad \text{if and only if} \quad x \in R_i$$

Encoder-Decoder structure

- *Encoder*: $\mathcal{E} : \mathbb{R} \rightarrow \{1, \dots, N\}$ defined by $\mathcal{E}(x) = j \Leftrightarrow x \in R_j$
- *Decoder*: $\mathcal{D} : \{1, \dots, N\} \rightarrow \{y_1, \dots, y_N\}$ defined by $\mathcal{D}(j) = y_j$
- Thus $Q(x) = \mathcal{D}(\mathcal{E}(x))$



- *Rate of Q :*

$$R(Q) = \log_2 N \quad \text{bits/sample}$$

- R is the number of bits needed to represent the N indices with binary words of a fixed length (fixed-rate binary encoding)

Definition (Regular quantizer)

Q is called a *regular* quantizer if

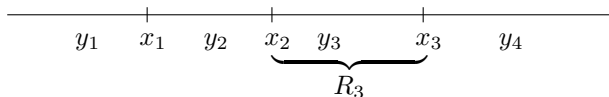
- (a) each R_i is an interval $(x_{i-1}, x_i]$ (or $[x_{i-1}, x_i)$ or (x_{i-1}, x_i) , etc).

Note: $x_{i-1} = \pm\infty$ and/or $x_i = \pm\infty$ are not excluded

- (b) $y_i \in R_i$ for all $i = 1, \dots, N$.

- $x_0, x_1, \dots, x_N$ are called the *decision levels*
- For a regular quantizer $x_0 < y_1 < x_1 < y_2 < \dots < y_N < x_N$.

Example: 4-level regular quantizer



Example: Infinite-level uniform quantizer

Define

$$Q(x) = \left\lfloor x + \frac{1}{2} \right\rfloor$$

- The output levels are $y_i = i$, $i = 0, \pm 1, \pm 2, \dots$, $N = \infty$.
- The codebook is $\mathcal{C} = \mathbb{Z}$, the set of all integers.
- The quantizer cells are determined as:

$$Q(x) = i \quad \Leftrightarrow \quad i \leq x + \frac{1}{2} < i + 1 \quad \Leftrightarrow \quad i - \frac{1}{2} \leq x < i + \frac{1}{2}$$

so

$$R_i = \left[i - \frac{1}{2}, i + \frac{1}{2} \right), \quad i = 0, \pm 1, \pm 2, \dots$$

Quality of reproduction

- *Distortion measure*: $d(x, y)$ measures the distortion (or loss) resulting if the input x is reproduced as y .
- Mathematically, we require $d(x, y) \geq 0$ for all $x, y \in \mathbb{R}$.
- Instantaneous distortion of Q for input x :

$$d(x, Q(x))$$

- A random variable (r.v.) X is often called a *random source*.
Expected distortion of Q :

$$D(Q) = Ed(X, Q(X))$$

To be useful, the distortion measure should be

- tractable (easy to compute)
- meaningful for perception or application

Some popular choices:

- $d(x, y) = (x - y)^2$ (squared error distortion, ubiquitous)
- $d(x, y) = |x - y|^r$ ($r > 0$; $r = 1$ is the most common choice)

Assume

- The source X has probability density function (pdf) f (short notation: $X \sim f$);
- Q is a regular quantizer with output levels $y_1, \dots, y_N$ and decision levels $x_0, \dots, x_N$.

Then the expected distortion is

$$\begin{aligned} Ed(X, Q(X)) &= \int_{-\infty}^{\infty} d(x, Q(x)) f(x) dx \\ &= \sum_{i=1}^N \int_{R_i} d(x, Q(x)) f(x) dx \\ &= \sum_{i=1}^N \int_{x_{i-1}}^{x_i} d(x, y_i) f(x) dx \end{aligned}$$

Example: N -level *uniform quantizer* over the interval $(a, b]$: Let $\Delta = (b - a)/N$ and for $i = 1, \dots, N$,

$$R_i = (a + (i - 1)\Delta, a + i\Delta]$$

$$y_i = a + \Delta \left(i - \frac{1}{2}\right)$$

Assume X is uniformly distributed over $[a, b]$:

$$f(x) = \begin{cases} \frac{1}{b - a} & \text{if } a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$

Consider the squared error $d(x, y) = (x - y)^2$.

Example cont'd:

$$\begin{aligned} Ed(X, Q(X)) &= \sum_{i=1}^N \int_{R_i} d(x, y_i) f(x) dx \\ &= \sum_{i=1}^N \int_{a+(i-1)\Delta}^{a+i\Delta} (x - y_i)^2 f(x) dx \\ &= \frac{1}{b-a} \sum_{i=1}^N \int_{a+(i-1)\Delta}^{a+i\Delta} (x - y_i)^2 dx \\ &= \frac{1}{b-a} N \int_{-\Delta/2}^{\Delta/2} z^2 dz \quad (\text{change of variable } z = x - y_i) \\ &= \frac{1}{\Delta} \int_{-\Delta/2}^{\Delta/2} z^2 dz \\ &= \frac{\Delta^2}{12} \end{aligned}$$

We conclude

$$D(Q) = \frac{\Delta^2}{12}$$

Signal-to-noise ratio

Assume $d(x, y) = (x - y)^2$. Then $D(Q)$ is called the *mean squared error* (MSE) of Q .

- $e = Q(X) - X$ is called the quantization error (noise) since

$$Q(X) = X + Q(X) - X = X + e$$

- $D(Q) = E[(X - Q(X))^2] = E(e^2)$
- *Signal-to-noise ratio* (signal-to-quantization noise ratio):

$$\text{SNR} = 10 \log_{10} \frac{E(X^2)}{D(Q)} \quad \text{dB}$$

Performance of quantizers

The performance of an N -level quantizer is measured by its distortion and rate:

$$D(Q) = Ed(X, Q(X))$$

$$R(Q) = \log_2 N$$

Goal: Find optimal tradeoffs between distortion and rate; i.e., minimize one while constraining the other.

Note: distortion and rate are *conflicting* quantities

Optimality of scalar quantizers

Definition

Let $\mathcal{Q}_N$ denote the family of *all* N -level quantizers. $Q^* \in \mathcal{Q}_N$ is an optimal quantizer if

$$Ed(X, Q^*(X)) = \min_{Q \in \mathcal{Q}_N} Ed(X, Q(X))$$

Remarks:

- Q^* depends on the distribution of X and on d .
- Q^* is not necessarily unique (there may be more than one optimal N -level quantizers).
- Can be shown that Q^* exists for all N and “reasonable” d if $Ed(X, y) < \infty$ for some $y \in \mathbb{R}$.

Theorem 1 (Nearest Neighbor Condition)

Consider all N -level scalar quantizers with codebook $\mathcal{C} = \{y_1, \dots, y_N\}$. Among these, any quantizer with quantization regions satisfying

$$R_i \subset \{x : d(x, y_i) \leq d(x, y_j), \quad j = 1, \dots, N\} \quad i = 1, \dots, N \quad (*)$$

has minimum distortion.

Note: Q with codebook $\mathcal{C}$ satisfies $(*)$ if and only if for any x ,

$$Q(x) = y_i \quad \text{implies} \quad d(x, y_i) \leq d(x, y_j) \quad \text{for all } j$$

Equivalently, for all x ,

$$d(x, Q(x)) = \min_{y_j \in \mathcal{C}} d(x, y_j)$$

or

$$Q(x) = \arg \min_{y_j \in \mathcal{C}} d(x, y_j)$$

“Proof” of Theorem 1: Let $Q \in \mathcal{Q}_N$ with codebook $\mathcal{C}$ satisfy $(*)$, and let $\hat{Q} \in \mathcal{Q}_N$ have codebook $\mathcal{C}$ but arbitrary quantization regions.

Then for all x ,

$$d(x, \hat{Q}(x)) \geq \min_{y_j \in \mathcal{C}} d(x, y_j) = d(x, Q(x))$$

Thus

$$Ed(X, \hat{Q}(X)) \geq Ed(X, Q(X))$$



Remarks:

- The subsets

$$\tilde{R}_i = \{x : d(x, y_i) \leq d(x, y_j), \quad j = 1, \dots, N\} \quad i = 1, \dots, N$$

cover the real line (i.e., $\bigcup_{i=1}^N \tilde{R}_i = \mathbb{R}$), but they are *not* disjoint.

- A simple tie-breaking rule giving an optimal partition is

$$R_1 = \tilde{R}_1, \quad R_i = \tilde{R}_i \setminus \bigcup_{j=1}^{i-1} \tilde{R}_j \quad \text{for } i = 2, \dots, N$$

Special Case: $d(x, y) = |x - y|^r$ (includes MSE)

$$|x - y_i|^r \leq |x - y_j|^r \quad \Leftrightarrow \quad |x - y_i| \leq |x - y_j|$$

Thus

$$\tilde{R}_i = \{x : |x - y_i| \leq |x - y_j|, \quad j = 1, \dots, N\} \quad i = 1, \dots, N$$

If $y_1 < y_2 < \dots < y_N$, then $\tilde{R}_i = [x_{i-1}, x_i]$, where

$$x_{i-1} = \frac{y_{i-1} + y_i}{2} \quad \text{and} \quad x_i = \frac{y_i + y_{i+1}}{2}, \quad i = 2, \dots, N - 1$$

To uniquely define R_i , we choose $R_i = (x_{i-1}, x_i]$, $i = 2, \dots, N - 1$.

Given $\mathcal{C}$, the optimal quantizer is *regular*. The decision levels are the *midpoints* between two neighboring output levels.

Theorem 2 (Centroid Condition)

Consider all N -level scalar quantizers with a given partition $\{R_1, \dots, R_N\}$. Among these, the quantizer Q with output levels

$$y_i = \arg \min_{y \in \mathbb{R}} E[d(X, y) | X \in R_i], \quad i = 1, \dots, N$$

has minimum distortion.

- y_i is called the *centroid* of R_i .
- The centroid y_i may not be unique.

Proof of Theorem 2: Let $P_i = P(X \in R_i)$. For any $\hat{Q}$ with partition $\{R_1, \dots, R_N\}$ and arbitrary output levels $\{\hat{y}_1, \dots, \hat{y}_N\}$,

$$\begin{aligned} Ed(X, \hat{Q}(X)) &= \sum_{i=1}^N E[d(X, \hat{Q}(X)) | X \in R_i] P_i \\ &= \sum_{i=1}^N E[d(X, \hat{y}_i) | X \in R_i] P_i \\ &\geq \sum_{i=1}^N \min_y E[d(X, y) | X \in R_i] P_i \\ &= \sum_{i=1}^N E[d(X, y_i) | X \in R_i] P_i = Ed(X, Q(X)) \quad \square \end{aligned}$$

Special Case: $d(x, y) = (x - y)^2$.

Theorem 3 (Centroid Condition for MSE)

The y_i 's minimizing the distortion given a partition $\{R_1, \dots, R_N\}$ are uniquely given by

$$y_i = E[X|X \in R_i], \quad i = 1, \dots, N$$

Note: If $X \sim f$, then

$$f_{X|R_i}(x) = \begin{cases} \frac{f(x)}{P(X \in R_i)} & \text{if } x \in R_i \\ 0 & \text{otherwise} \end{cases}$$

so

$$E[X|X \in R_i] = \int_{-\infty}^{\infty} x f_{X|R_i}(x) dx = \frac{\int_{R_i} x f(x) dx}{\int_{R_i} f(x) dx}$$

Proof of Theorem 3: Recall that for any r.v. Z with $E(Z^2) < \infty$

$$E[(Z - E(Z))^2] \leq E[(Z - y)^2] \quad (*)$$

and equality holds iff $y = E(Z)$.

Let Z_i have the conditional distribution of X given $X \in R_i$. Then

$$E(Z_i) = E[X|X \in R_i] = y_i$$

For any y

$$\begin{aligned} E[(X - y)^2|X \in R_i] &= E[(Z_i - y)^2] \\ &\geq E[(Z_i - E(Z_i))^2] \quad \text{from } (*) \\ &= E[(Z_i - y_i)^2] \\ &= E[(X - y_i)^2|X \in R_i] \end{aligned}$$

Also, y_i is the unique minimizer by $(*)$. □

Remarks:

- The Nearest Neighbor Condition (NNC) and the Centroid Condition (CC) are *necessary conditions for optimality*.
- An optimal quantizer has to satisfy both the NNC and the CC. The converse is not true: NNC+CC *do not* imply that a quantizer is optimal.
- NNC+CC = *Lloyd-Max conditions*. A quantizer satisfying both conditions is called a *Lloyd-Max quantizer*.
- The NNC determines the *optimal encoder* $\mathcal{E}$ given a decoder $\mathcal{D}$. The CC determines the optimal decoder $\mathcal{D}$ given an encoder $\mathcal{E}$.
- The NNC is *independent* of the source distribution (depends only on the distortion measure).
- The NNC implies that an optimal quantizer can be described by its *codebook* alone.

Simple examples for Lloyd-Max quantizers (MSE)

Example 1 (2-level quantizer): Assume $X \sim f$ and that $f(x) = f(-x)$ for all x . Let

$$c = E[X|X > 0] = \frac{\int_0^\infty x f(x) dx}{\int_0^\infty f(x) dx} = 2 \int_0^\infty x f(x) dx$$

and define

$$Q(x) = \begin{cases} c & \text{if } x > 0 \\ -c & \text{if } x \leq 0 \end{cases}$$

Then $N = 2$, $R_1 = (-\infty, 0]$, $y_1 = -c$, $R_2 = (0, \infty)$, $y_2 = c$.

Check NNC: $x_1 = \frac{y_1 + y_2}{2} = 0$ (satisfied)

Check CC: $y_i = E[X|X \in R_i]$, $i = 1, 2$ (satisfied)

Note: Q is a Lloyd-Max quantizer, but not necessarily optimal!

Example cont'd: Special case: assume $X \sim N(0, \sigma^2)$

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-x^2/(2\sigma^2)}$$

Then

$$\begin{aligned} c &= \frac{2}{\sqrt{2\pi}\sigma} \int_0^\infty x e^{-x^2/(2\sigma^2)} dx &= \frac{2}{\sqrt{2\pi}\sigma} \left[-\sigma^2 e^{-x^2/(2\sigma^2)} \right]_0^\infty \\ & &= \sigma \sqrt{\frac{2}{\pi}} \end{aligned}$$

It can be proved that Q is indeed the (unique) optimal 2-level quantizer for the Gaussian case.

Example 2 (Uniform quantizer): N -level *uniform quantizer* over the interval $(a, b]$: for $i = 1, \dots, N$

$$\begin{aligned} R_i &= (a + (i - 1)\Delta, a + i\Delta] \\ y_i &= a + \Delta \left(i - \frac{1}{2}\right) \end{aligned}$$

- Each R_i is an interval of length $\Delta = \frac{b - a}{N}$
- y_i is the midpoint of R_i .
- $x_i = \frac{y_i + y_{i+1}}{2} \implies$ **NNC is satisfied.**

Assume X is uniformly distributed over $[a, b]$:

$$f(x) = \begin{cases} \frac{1}{b - a} & \text{if } a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$

Example cont'd: We have

$$f_{X|R_i}(x) = \begin{cases} \frac{f(x)}{P(X \in R_i)} = \frac{1}{\Delta} & \text{if } x \in R_i \\ 0 & \text{otherwise} \end{cases}$$

Thus, given $X \in R_i$, the distribution of X is *uniform* on $R_i = (a + (i - 1)\Delta, a + i\Delta]$ and we get

$$E[X|X \in R_i] = a + \Delta \left(i - \frac{1}{2}\right) = y_i \quad (\text{midpoint of } R_i)$$

$\implies$ CC is satisfied.

The Lloyd quantizer design algorithm

Idea: For a fixed codebook, optimize the partition. For the resulting partition, optimize the codebook. Iterate...

Lloyd Iteration

- (a) Given $\mathcal{C}_m = \{y_1^{(m)}, \dots, y_N^{(m)}\}$, use the NN condition to form the optimal partition

$$R_i^{(m)} = \{x : d(x, y_i^{(m)}) \leq d(x, y_j^{(m)}), \quad j = 1, \dots, N\} \quad i = 1, \dots, N$$

- (b) Determine $\mathcal{C}_{m+1} = \{y_1^{(m+1)}, \dots, y_N^{(m+1)}\}$ using the centroid condition

$$y_i^{(m+1)} = \arg \min_{y \in \mathbb{R}} E[d(X, y) | X \in R_i^{(m)}], \quad i = 1, \dots, N$$

$$(\text{For MSE } y_i^{(m+1)} = E[X | X \in R_i^{(m)}])$$

Lloyd algorithm

- Step 1** Inputs: pdf $f(x)$, initial codebook $\mathcal{C}_1 = \{y_1^{(1)}, \dots, y_N^{(1)}\}$, threshold $\epsilon > 0$. Set $m = 1$ and $D_1 = Ed(X, Q^{(1)}(X))$.
- Step 2** Given $\mathcal{C}_m$, perform the Lloyd iteration (a) and (b) to generate $\mathcal{C}_{m+1}$.
- Step 3** Compute $D_{m+1} = Ed(X, Q^{(m+1)}(X))$.
If $\frac{D_m - D_{m+1}}{D_m} < \epsilon$, then output $\mathcal{C}_{m+1}$ and stop.
Otherwise $m := m + 1$ and go to **Step 2**.

Remarks:

- NNC+CC: (a) and (b) in the Lloyd iteration either reduce the distortion or leave it unchanged. Thus

$$D_1 \geq D_2 \geq \cdots \geq D_m \geq D_{m+1} \geq \cdots$$

and D_m converges to a *limit* as $m \rightarrow \infty$. Hence

$$\lim_{m \rightarrow \infty} \frac{D_m - D_{m+1}}{D_m} = 0$$

and so the algorithm always stops after a finite number of iterations.

- It is not guaranteed that the sequence of codebooks $\mathcal{C}_m$ converges as $m \rightarrow \infty$
- Even if $\mathcal{C}_m$ converges, the “limit quantizer” may not be optimal (it is only guaranteed to be “locally optimal”)

Design from training data

In practice, the distribution (pdf) of X is unknown. Instead, independent (or dependent) samples $V_1, V_2, \dots, V_M$ with $V_j \sim X$ are given.

- (i) We could form an estimate $f_M(x)$ of $f(x)$ from $V_1, \dots, V_M$. Then use the Lloyd algorithm with f_M . Computationally demanding.
- (ii) *Direct approach*: given the training set $\{v_1, \dots, v_M\}$, approximate the distribution of X by the distribution of a *discrete* r.v.

$$P(Z_M = v_j) = \frac{1}{M}$$

i.e., assign probability mass $1/M$ to each v_j .

If the v_j are all distinct (which happens with prob. 1 if produced by a source with a pdf), then a valid discrete distribution is obtained.

This is the *empirical distribution* of the training sequence.

Remarks:

- For any quantizer Q

$$\begin{aligned} Ed(Z_M, Q(Z_M)) &= \sum_{j=1}^M d(v_j, Q(v_j)) P(Z_M = v_j) \\ &= \frac{1}{M} \sum_{j=1}^M d(v_j, Q(v_j)) \end{aligned}$$

Thus optimizing Q for Z_M is *equivalent* to minimizing the *average* distortion of Q over the *training set*.

- Assume $v_1, \dots, v_M$ are drawn independently from the distribution of X or form a stationary and ergodic sequence. It can be shown that as $M \rightarrow \infty$, the distribution of Z_M (the empirical distribution) approximates the distribution of X increasingly well. Thus for large M , minimizing the *average* distortion of Q over the training set approximates optimizing Q for X .

Lloyd-Max conditions for training sets (MSE)

NN condition: Since independent of the source, remains unchanged.

Need only to partition the training set $\mathcal{T} = \{v_1, \dots, v_M\}$:

$$R_i = \{v \in \mathcal{T} : |v - y_i| \leq |v - y_j|, j = 1, \dots, N\} \quad i = 1, \dots, N$$

Tie breaking: if $v \in \mathcal{T}$ has two nearest neighbors, assign it to the one with the smaller index.

Centroid condition: Given $R_1, \dots, R_N$, the MSE centroids are

$$\begin{aligned} y_i = E[Z_M | Z_M \in R_i] &= \sum_{j=1}^M v_j P(Z_M = v_j | Z_M \in R_i) \\ &= \frac{1}{|R_i|} \sum_{v_j \in R_i} v_j \end{aligned}$$

where $|R_i|$ is the number of training samples in R_i .

Lloyd algorithm for training data and MSE

Step 1 Inputs: training set $\mathcal{T} = \{v_1, \dots, v_M\}$, initial codebook

$\mathcal{C}_1 = \{y_1^{(1)}, \dots, y_N^{(1)}\}$, and threshold $\epsilon > 0$. Set $m = 1$.

Step 2 Given $\mathcal{C}_m$, partition $\mathcal{T}$ into N bins $R_1^{(m)}, \dots, R_N^{(m)}$ using the NNC. Calculate the empirical average inside each bin

$$y_i^{(m+1)} = \frac{1}{|R_i^{(m)}|} \sum_{v_j \in R_i^{(m)}} v_j$$

to generate $\mathcal{C}_{m+1}$.

Step 3 Compute

$$D_{m+1} = \frac{1}{M} \sum_{j=1}^M (v_j - Q(v_j))^2$$

If $(D_m - D_{m+1})/D_m < \epsilon$, then output $\mathcal{C}_{m+1}$ and stop. Otherwise $m := m + 1$ and go to **Step 2**.

Performance Analysis

Distortion of optimal N -level quantizer:

$$D^*(N) = \min_{Q \in \mathcal{Q}_N} Ed(X, Q(X))$$

For $X \sim f$ and MSE

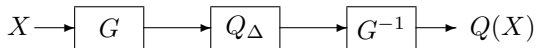
$$\begin{aligned} D^*(N) &= \min_{Q \in \mathcal{Q}_N} \int_{-\infty}^{\infty} (x - Q(x))^2 f(x) dx \\ &= \min_{y_1 < y_2 < \dots < y_N} \underbrace{\sum_{i=1}^N \int_{x_{i-1}}^{x_i} (x - y_i)^2 f(x) dx}_{g(y_1, y_2, \dots, y_N)} \end{aligned}$$

where $x_i = \frac{1}{2}(y_i + y_{i+1})$, $i = 1, \dots, N - 1$.

Thus determining $D^*(N)$ and the optimal Q^* involves the minimization of a real function of N real variables. *Very hard, nonlinear and non-convex problem!*

Companding quantization

Provides structure facilitating performance analysis



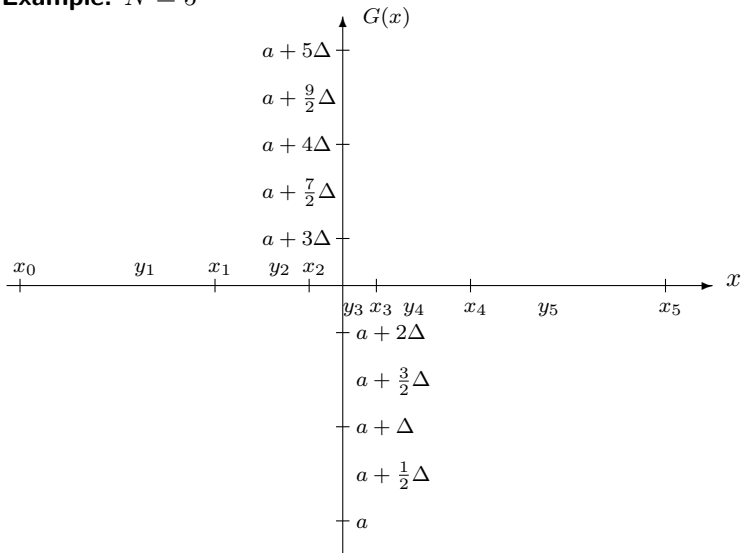
- $G : \mathbb{R} \rightarrow (a, b)$ monotone increasing and invertible (compressor)
- Q_{Δ} : N -level uniform quantizer with support (a, b) and step size $\Delta = \frac{b-a}{N}$.
- $G^{-1} : (a, b) \rightarrow \mathbb{R}$, the inverse of G (expandor)

$$Q(x) = G^{-1}(Q_{\Delta}(G(x)))$$

Proposition 1

For any N -level *regular* quantizer Q there exists a companding realization.

Example: $N = 5$



Proof of Proposition 1: For simplicity assume Q has bounded support. The output levels and decision levels are ordered as $x_0 < y_1 < x_1 < y_2 < \cdots < y_N < x_N$.

Prescribe the values of G such that

- $G(y_i) = a + \Delta(i - \frac{1}{2}), \quad i = 1, \dots, N$
- $G(x_i) = a + i\Delta, \quad i = 0, \dots, N$
- $G(x)$ is *linearly* interpolated between these discrete points

Then

- $G(x) : (x_0, x_N) \rightarrow (a, b)$ is strictly monotone increasing (so it is invertible)
- $x \in (x_{i-1}, x_i] \Rightarrow G(x) \in (a + (i-1)\Delta, a + i\Delta] \Rightarrow$
 $\Rightarrow Q_\Delta(G(x)) = a + \Delta(i - \frac{1}{2}) \Rightarrow G^{-1}(Q_\Delta(G(x))) = y_i$

□

Note: $G(x)$ can be “smoothed out” while retaining these properties.

Another look at $D^*(N)$

- Fix (a, b) and let $Q_{G,N}$ denote the N -level quantizer realized by the compressor function G .
- By the proposition

$$D^*(N) = \min_{Q \in \mathcal{Q}_N} E[(X - Q(X))^2] = \min_G E[(X - Q_{G,N}(X))^2]$$

- Hence it is of great interest to determine $\min_G E[(X - Q_{G,N}(X))^2]$
- Will do this under the assumption that N is large (“high-resolution conditions”)

High-resolution performance

Fix G , let $N \rightarrow \infty$

Assumptions

- (i) N is large, $\Delta_i = (x_i - x_{i-1})$ is small for $i = 2, \dots, N - 1$
- (ii) $X \sim f$, where f is a continuous pdf
- (iii) $G : \mathbb{R} \rightarrow (a, b)$ is continuously differentiable

Will do a series of approximations based on these assumptions

$$\begin{aligned}
D(Q_{G,N}) &= \sum_{i=1}^N \int_{x_{i-1}}^{x_i} (x - y_i)^2 f(x) dx \\
&= \underbrace{\sum_{i=2}^{N-1} \int_{x_{i-1}}^{x_i} (x - y_i)^2 f(x) dx}_{\text{"granular" distortion } D_{\text{gr}}} \\
&\quad + \underbrace{\int_{-\infty}^{x_1} (x - y_1)^2 f(x) dx + \int_{x_{N-1}}^{\infty} (x - y_N)^2 f(x) dx}_{\text{"overload" distortion } D_{\text{ol}}}
\end{aligned}$$

Assumption (iv): The overload distortion is *negligible* compared to the granular distortion

$$D(Q) \approx D_{\text{gr}}$$

Consider cell $R_i = (x_{i-1}, x_i]$.

$a + i\Delta = G(x_i)$ implies $G(x_i) - G(x_{i-1}) = \Delta$. Since $x_i - x_{i-1}$ is assumed to be small

$$\frac{\Delta}{x_i - x_{i-1}} = \frac{G(x_i) - G(x_{i-1})}{x_i - x_{i-1}} \approx G'(y_i)$$

(recall $y_i \in (x_{i-1}, x_i]$). Thus

$$(x_i - x_{i-1}) \approx \frac{\Delta}{G'(y_i)}$$

A similar argument shows

$$y_i \approx \frac{x_{i-1} + x_i}{2}$$

Thus

$$\begin{aligned}\int_{x_{i-1}}^{x_i} (x - y_i)^2 f(x) dx &\approx f(y_i) \int_{x_{i-1}}^{x_i} (x - y_i)^2 dx \\ &\quad \text{(since } f \text{ is continuous and } x_i - x_{i-1} \text{ is small)} \\ &\approx f(y_i) \int_{x_{i-1}}^{x_i} \left(x - \frac{x_{i-1} + x_i}{2}\right)^2 dx \\ &= f(y_i) \frac{(x_i - x_{i-1})^3}{12} \\ &\approx \frac{1}{12} f(y_i) \frac{\Delta^2}{G'(y_i)^2} (x_i - x_{i-1}) \\ &= \frac{1}{12} f(y_i) \frac{B^2}{N^2} \frac{1}{G'(y_i)^2} (x_i - x_{i-1}) \\ &\quad \text{(by letting } B = (b - a) \text{ so that } \Delta = \frac{B}{N} \text{)}\end{aligned}$$

Putting things together

$$\begin{aligned} D(Q_{G,N}) &\approx \sum_{i=2}^{N-1} \int_{x_{i-1}}^{x_i} (x - y_i)^2 f(x) dx \\ &\approx \frac{B^2}{12N^2} \sum_{i=2}^{N-1} \frac{f(y_i)}{G'(y_i)^2} (x_i - x_{i-1}) \\ &\hspace{15em} \text{(Riemann sum)} \\ &\approx \frac{B^2}{12N^2} \int_{x_1}^{x_{N-1}} \frac{f(x)}{G'(x)^2} dx \\ &\approx \boxed{\frac{B^2}{12N^2} \int_{-\infty}^{\infty} \frac{f(x)}{G'(x)^2} dx} \end{aligned}$$

Conclusion: For MSE and N large

$$D(Q_{G,N}) \approx \frac{B^2}{12N^2} \int_{-\infty}^{\infty} \frac{f(x)}{G'(x)^2} dx$$

(Here $\approx$ means that the ratio of the two sides converges to 1 as $N \rightarrow \infty$)

Easy-to-calculate, analytic formula!

Example: Let $X \sim f$ such that $f(x) = 0$ if $x \notin (a, b)$. Consider the N -level uniform quantizer Q_Δ over (a, b) , where $\Delta = (b - a)/N$.

If $G : (a, b) \rightarrow (a, b)$ is defined as $G(x) = x$, then for all $x \in (a, b)$

$$Q_{N,G}(x) = G^{-1}(Q_\Delta(G(x))) = Q_\Delta(x)$$

From the compander approximation, for large N

$$\begin{aligned} D(Q_\Delta) = D(Q_{G,N}) &\approx \frac{(b-a)^2}{12N^2} \int_{-\infty}^{\infty} \frac{f(x)}{G'(x)^2} dx \\ &= \frac{1}{12} \left(\frac{b-a}{N} \right)^2 \int_a^b f(x) dx \quad (\text{since } G'(x) = 1) \\ &= \boxed{\frac{\Delta^2}{12}} \end{aligned}$$

Intuitive explanation: If N is large, Δ is small, and the pdf is approximately *uniform* over each quantizer cell

Optimal asymptotic performance

- We want to find the compressor G minimizing $D(Q_{G,N})$ for large N .
- Any N -level quantizer can be represented as a companding quantizer with compressor function $G : \mathbb{R} \rightarrow (0, 1)$.
 $\Rightarrow$ It is enough to consider compressors $G : \mathbb{R} \rightarrow (0, 1)$.
- In this case (since $B = 1$)

$$D(Q_{G,N}) \approx \frac{1}{12N^2} \int_{-\infty}^{\infty} \frac{f(x)}{G'(x)^2} dx$$

- We want to minimize

$$\int_{-\infty}^{\infty} \frac{f(x)}{G'(x)^2} dx$$

Theorem 4 (Optimal compressor)

Given the pdf f , the compressor $G : \mathbb{R} \rightarrow (0, 1)$ which minimizes

$$\int_{-\infty}^{\infty} \frac{f(x)}{G'(x)^2} dx$$

is determined by

$$G'(x) = \frac{f(x)^{1/3}}{\int_{-\infty}^{\infty} f(y)^{1/3} dy}$$

For the (asymptotically) optimal companding scheme using this G

$$D(Q_{G,N}) \approx \frac{1}{12N^2} \left(\int_{-\infty}^{\infty} f(x)^{1/3} dx \right)^3$$

Lemma 5 (Hölder's inequality)

Assume $p > 1$ and $q > 1$ are such that $\frac{1}{p} + \frac{1}{q} = 1$. Let $u(x) \geq 0$ and $v(x) \geq 0$ satisfy

$$\int_{-\infty}^{\infty} u(x)^p dx < \infty \quad \int_{-\infty}^{\infty} v(x)^q dx < \infty$$

Then

$$\int_{-\infty}^{\infty} u(x)v(x) dx \leq \left(\int_{-\infty}^{\infty} u(x)^p dx \right)^{1/p} \left(\int_{-\infty}^{\infty} v(x)^q dx \right)^{1/q}$$

Moreover, equality holds iff $v(x)^q = Cu(x)^p$ for some $C > 0$.

Remark: For any $p > 0$ and function $g : \mathbb{R} \rightarrow \mathbb{R}$ define

$$\|g\|_p = \left(\int_{-\infty}^{\infty} |g(x)|^p dx \right)^{1/p}$$

Then Hölder's inequality states that

$$\int_{-\infty}^{\infty} u(x)v(x) dx \leq \|u\|_p \|v\|_q$$

Proof of Lemma 5: First we note that the strict concavity of $\log(x)$ implies that for any $a, b \geq 0$ and $0 < \alpha < 1$,

$$a^\alpha b^{1-\alpha} \leq \alpha a + (1 - \alpha)b \quad (*)$$

where equality holds iff $a = b$. (Proof: if $ab > 0$, take the logarithm of both sides.)

Define

$$a(x) = \left(\frac{u(x)}{\|u\|_p} \right)^p, \quad b(x) = \left(\frac{v(x)}{\|v\|_q} \right)^q$$

Let $\alpha = \frac{1}{p}$ (so that $1 - \alpha = \frac{1}{q}$). From $(*)$ we have

$$\begin{aligned} \frac{u(x)}{\|u\|_p} \cdot \frac{v(x)}{\|v\|_q} &= a(x)^\alpha b(x)^{1-\alpha} \\ &\leq \alpha a(x) + (1 - \alpha)b(x) \\ &= \alpha \left(\frac{u(x)}{\|u\|_p} \right)^p + (1 - \alpha) \left(\frac{v(x)}{\|v\|_q} \right)^q \end{aligned}$$

Now integrate both sides.

Proof of Lemma cont'd:

$$\begin{aligned} \int_{-\infty}^{\infty} \frac{u(x)}{\|u\|_p} \cdot \frac{v(x)}{\|v\|_q} dx \\ \leq \underbrace{\alpha \int_{-\infty}^{\infty} \left(\frac{u(x)}{\|u\|_p} \right)^p dx}_{=1} + (1-\alpha) \underbrace{\int_{-\infty}^{\infty} \left(\frac{v(x)}{\|v\|_q} \right)^q dx}_{=1} \\ = \alpha + (1-\alpha) = 1 \end{aligned}$$

Hence

$$\int_{-\infty}^{\infty} u(x)v(x) dx \leq \|u\|_p \|v\|_q$$

Note that equality holds if and only if $a(x) = b(x)$ for (almost) all x , i.e.,
iff

$$\left(\frac{u(x)}{\|u\|_p} \right)^p = \left(\frac{v(x)}{\|v\|_q} \right)^q$$

This is equivalent to $v(x)^q = Cu(x)^p$ for some $C > 0$. □

Proof of Thm. 4: Let $G : \mathbb{R} \rightarrow (0, 1)$ be an arbitrary compressor. Define

$$u(x) = \left(\frac{f(x)}{G'(x)^2} \right)^{1/3} \quad v(x) = G'(x)^{2/3}$$

and let $p = 3$, $q = 3/2$ (so that $\frac{1}{p} + \frac{1}{q} = 1$).

By Hölder's inequality

$$\int_{-\infty}^{\infty} \underbrace{\left(\frac{f(x)}{G'(x)^2} \right)^{1/3} G'(x)^{2/3}}_{f(x)^{1/3}} dx \leq \left(\int_{-\infty}^{\infty} \frac{f(x)}{G'(x)^2} dx \right)^{1/3} \underbrace{\left(\int_{-\infty}^{\infty} G'(x) dx \right)^{2/3}}_{=1} \quad (*)$$

(*) follows since G is monotone increasing and maps onto $(0, 1)$ so

$$\begin{aligned} \int_{-\infty}^{\infty} G'(x) dx &= \lim_{y \rightarrow \infty} [G(x)]_{-y}^y = \lim_{y \rightarrow \infty} G(y) - \lim_{y \rightarrow -\infty} G(y) \\ &= 1 - 0 = 1 \end{aligned}$$

Proof of Thm. 4 cont'd:

We showed

$$\int_{-\infty}^{\infty} \frac{f(x)}{G'(x)^2} dx \geq \left(\int_{-\infty}^{\infty} f(x)^{1/3} dx \right)^3 \quad (**)$$

From Lemma 5, the lower bound (**) is achieved by G iff

$$G'(x) = C \frac{f(x)}{G'(x)^2} \iff G'(x) = \hat{C} f(x)^{1/3}$$

Since $\int_{-\infty}^{\infty} G'(x) dx = 1$, we get $\hat{C} = \left(\int_{-\infty}^{\infty} f(x)^{1/3} dx \right)^{-1}$

Thus the optimal compressor satisfies

$$G'(x) = \frac{f(x)^{1/3}}{\int_{-\infty}^{\infty} f(y)^{1/3} dy}$$

□

We have “proved” that for continuous sources with a pdf and N large

$$D^*(N) = \min_G D(Q_{G,N}) \approx \frac{1}{12N^2} \|f\|_{1/3}$$

The exact result (which is much harder to prove) is

Theorem 6 (Bucklew and Wise)

Assume the source pdf satisfies $\int_{-\infty}^{\infty} |x|^{2+\epsilon} f(x) dx < \infty$ for some $\epsilon > 0$.

Then

$$\lim_{N \rightarrow \infty} N^2 D^*(N) = \frac{1}{12} \|f\|_{1/3}$$

The 6 dB per bit rule

- Consider the SNR as a function of $R = \log_2 N$. Since $N^{-2} = 2^{-2R}$

$$D^*(N) \approx \frac{1}{N^2} \frac{1}{12} \|f\|_{1/3} = 2^{-2R} C(f)$$

where $C(f) = \frac{1}{12} \|f\|_{1/3}$ does not depend on N .

- Thus the SNR in dB is

$$\begin{aligned} \text{SNR} &\approx 10 \log_{10} \left(\frac{E(X^2)}{2^{-2R} C(f)} \right) = R \cdot \underbrace{20 \log_{10} 2}_{\approx 6.02} + 10 \log_{10} \left(\frac{E(X^2)}{C(f)} \right) \\ &\approx \boxed{6R + \text{constant dB}} \end{aligned}$$

- Rule of thumb:* increasing the rate by 1 bit results in a 6 dB increase in SNR.